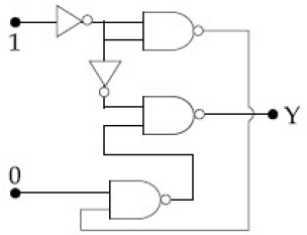
JEE Main Online Exam 2020

Questions & Solutions

8th January 2020 | Shift - II

PHYSICS

Q.1 In the given circuit, value of Y is :



(1) will not execute

(2) toggles between 0 and 1

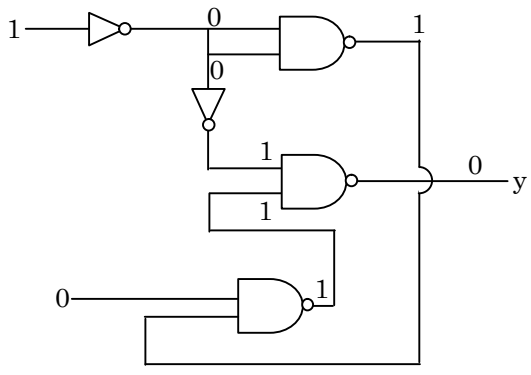
(3) 1

(4) 0

Ans. **[4]**

Sol. NAND Gate

x	y	x NAND y
0	0	1
0	1	1
1	0	1
1	1	0



y = 0

Q.2 A Carnot engine having an efficiency of $\frac{1}{10}$ is being used as a refrigerator. If the work done on the refrigerator is 10 J, the amount of heat absorbed from the reservoir at lower temperature is :

(1) 1J (2) 90 J

(3) 100 J (4) 99 J

Ans. **[2]**

Sol. $n = 1 - \frac{T_C}{T_H} = \frac{1}{10}$

$\therefore \frac{T_C}{T_H} = 1 - \frac{1}{10} = \frac{9}{10}$

$\therefore \frac{Q_C}{Q_H} = \frac{9}{10}$

$\Rightarrow Q_H = \frac{10}{9} Q_C$

Given : $Q_C + W = Q_H$

$\Rightarrow Q_C + 10 = \frac{10}{9} Q_C$

$\Rightarrow \frac{Q_C}{9} = 10$

$\Rightarrow Q_C = 90$

Q.3 A uniform sphere of mass 500 g rolls without slipping on a plane horizontal surface with its centre moving at a speed of 5.00 cm/s. Its kinetic energy is :

(1) 1.13×10^{-3} J (2) 8.75×10^{-3} J

(3) 6.25×10^{-4} J (4) 8.75×10^{-4} J

Ans. **[4]**

Sol. $K.E. = \frac{1}{2} mV^2 + \frac{1}{2} I_{cm} \omega^2$

$= \frac{1}{2} mV^2 + \frac{1}{2} \cdot \frac{2}{5} mR^2 \cdot \frac{V^2}{R^2}$

$= \frac{1}{2} mV^2 + \frac{1}{5} mV^2$

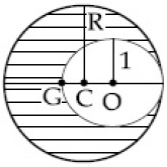
$= \frac{7}{10} mV^2$

$= \frac{7}{10} \times 0.5 \times 25 \times 10^{-4}$

$= \frac{35}{4} \times 10^{-4}$

$= 8.75 \times 10^{-4}$ J

Q.4 As shown in fig. when a spherical cavity (centred at O) of radius 1 is cut of a uniform sphere of radius R (centred at C), the centre of mass of remaining (shaded part of sphere is at G, i.e on the surface of the cavity. R can be determined by the equation :



(1) $(R^2 - R + 1)(2 - R) = 1$
(2) $(R^2 + R + 1)(2 - R) = 1$
(3) $(R^2 - R - 1)(2 - R) = 1$
(4) $(R^2 + R - 1)(2 - R) = 1$

Ans. **[2]**

Sol. $M_1 = \frac{4}{3}\pi R^3 . \rho$

$M_2 = \frac{4}{3}\pi(1^3) \times (-\rho)$

$X_G = \frac{M_1 X_C + M_2 X_0}{M_1 + M_2}$

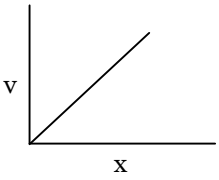
$OC = R - 1$

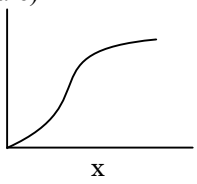
$CG = OG - OC = 1 - (R - 1) = 2 - R$

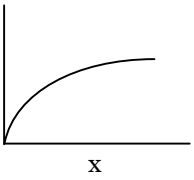
$$\Rightarrow -(2 - R) = \frac{M_1 \times 0 - \frac{4}{3}\pi\rho \cdot (R-1)}{\frac{4}{3}\pi R^3 \cdot \rho - \frac{4}{3}\pi\rho} \Rightarrow -(2 - R) = \frac{-\rho(R-1)}{\rho(R^3 - 1)}$$

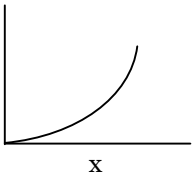
$$\Rightarrow (R^2 + R + 1)(2 - R) = 1$$

Q.5 A particle of mass m and charge q is released from rest in a uniform electric field. If there is no other force on the particle the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale)

(1) 

(2) 

(3) 

(4) 

Ans. **[3]**

Sol. $F = qE$

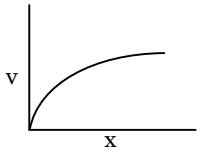
$a = \frac{F}{m} = \frac{qE}{m}$

$\frac{vdv}{dx} = \frac{qE}{m}$

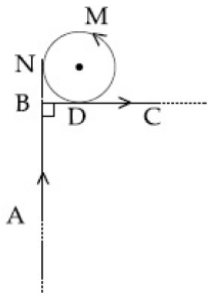
$\int_0^v vdv = \int_0^x \frac{qE}{m} dx$

$\frac{v^2}{2} = \frac{qE}{m} x$

$v = \sqrt{\frac{2qE}{m}}\sqrt{x}$



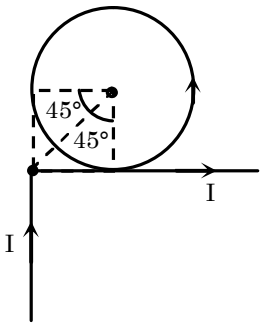
Q.6 A very long wire ABDMNDC is shown in figure carrying current I. AB and BC parts are straight, long and at right angle. At D wire forms a circular turn DMND of radius R. AB, BC parts are tangential to circular turn at N and D. Magnetic field at the centre of circle is :

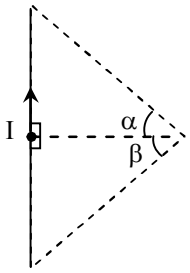


(1) $\frac{\mu_0 I}{2\pi R}\left(\pi - \frac{1}{\sqrt{2}}\right)$ (2) $\frac{\mu_0 I}{2\pi R}(\pi + 1)$ (3) $\frac{\mu_0 I}{2\pi R}\left(\pi + \frac{1}{\sqrt{2}}\right)$ (4) $\frac{\mu_0 I}{2R}$

Ans. **[3]**

Sol.





$$B = \frac{\mu_0 I}{4\pi d}(\sin\alpha + \sin\beta)$$

$$\overline{B} = \frac{\mu_0 I}{4\pi R}(\sin 90° - \sin 45°) \otimes$$

$$+ \frac{\mu_0 I}{2R} \odot$$

$$+ \frac{\mu_0 I}{4\pi R}(\sin 90° + \sin 45°) \odot$$

$$= \frac{\mu_0 I}{2\pi R}\left(\frac{1}{\sqrt{2}} + \pi\right) \odot$$

Q.7 Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2, respectively. The electric fields E_1 (on S_1) and E_2 (on S_2) on their surfaces are such that $E_1/E_2 = R_1/R_2$. Then the ratio V_1(on S_1)/ V_2(on S_2) of the electrostatic potentials on each sphere is :

(1) $\left(\frac{R_2}{R_1}\right)$ (2) $\left(\frac{R_1}{R_2}\right)^3$ (3) $\left(\frac{R_1}{R_2}\right)^2$ (4) $\frac{R_1}{R_2}$

Ans. **[3]**

Sol. $\frac{E_1}{E_2} = \frac{R_1}{R_2}$

$$\frac{V_1}{V_2} = \frac{E_1R_1}{E_2R_2} = \left(\frac{R_1}{R_2}\right)^2$$

Q.8 Consider a mixture of a n moles of helium gas and 2n moles of oxygen gas (molecules taken to be rigid) as an ideal gas. Its $\frac{C_P}{C_V}$ value will be :

(1) $\frac{19}{13}$ (2) $\frac{23}{15}$ (3) $\frac{40}{27}$ (4) $\frac{67}{45}$

Ans. **[1]**

Sol.

	C_V	C_P	
He	$\frac{3R}{2}$	$\frac{5R}{2}$	$\longrightarrow$ n
O_2	$\frac{5R}{2}$	$\frac{7R}{2}$	$\longrightarrow$ 2n

$$\frac{C_P}{C_V} = \frac{n_1C_{P_1} + n_2C_{P_2}}{n_1C_{V_1} + n_2C_{V_2}}$$

$$= \frac{n \times \frac{5R}{2} + 2n \times \frac{7R}{2}}{n \times \frac{3R}{2} + 2n \times \frac{5R}{2}}$$

$$= \frac{5+14}{3+10} = \frac{19}{13}$$

Q.9 A particle moves such that its position vector $\vec{r}(t) = \cos \omega t\, \hat{i} + \sin \omega t\, \hat{j}$ where ω is a constant and t is time. Then which of the following statements is true for the velocity $\vec{v}(t)$ and acceleration $\vec{a}(t)$ of the particle :

(1) $\vec{v}$ and $\vec{a}$ both are perpendicular to $\vec{r}$

(2) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed towards the origin

(3) $\vec{v}$ and $\vec{a}$ both are parallel to $\vec{r}$

(4) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed away from the origin

Ans. **[2]**

Sol. $\vec{r} = \cos\omega t\ \hat{i} + \sin\omega t\ \hat{j}$

$$\vec{v} = \frac{d\vec{r}}{dt} = -\omega \sin\omega t\ \hat{i} + \omega\cos\omega t\ \hat{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -\omega^2\cos\omega t\ \hat{i} - \omega^2 \sin\omega t\ \hat{j}$$

$$= -\omega^2(\cos\omega t\ \hat{i} + \sin\omega t\ \hat{j})$$

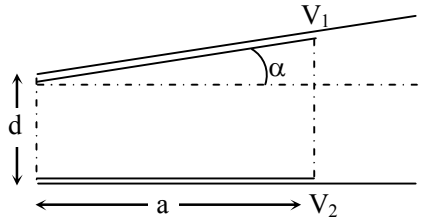
$$= -\omega^2\ \vec{r}$$

$\vec{v}\cdot\vec{r} = 0 \Rightarrow \vec{v} \perp \vec{r}$

Since, $\vec{a} = -\omega^2\ \vec{r}$

It is towards the origin

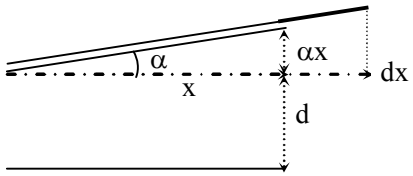
Q.10 A capacitor is made of two square plates each of side 'a' making a very small angles α between them, as shown in figure, The capacitance will be close to :



(1) $\frac{\epsilon_0 a^2}{d}\left(1-\frac{\alpha a}{2d}\right)$ (2) $\frac{\epsilon_0 a^2}{d}\left(1+\frac{\alpha a}{d}\right)$ (3) $\frac{\epsilon_0 a^2}{d}\left(1-\frac{\alpha a}{4d}\right)$ (4) $\frac{\epsilon_0 a^2}{d}\left(1-\frac{3\alpha a}{2d}\right)$

Ans. **[1]**

Sol.



$$dC = \frac{\epsilon_0\ a dx}{d+\alpha x}$$

$$C = \int dC = \frac{\epsilon_0\ a}{\alpha}\left[\ell n(\alpha x + d)\right]_0^a$$

$$= \frac{\epsilon_0\ a}{\alpha}\ \ell n\left(\frac{\alpha a + d}{d}\right) = \frac{\epsilon_0\ a}{\alpha}\ \ell n\left(1+\frac{\alpha a}{d}\right)$$

$$\approx \frac{\epsilon_0\ a}{\alpha}\left[\frac{\alpha a}{d} - \frac{\alpha^2 a^2}{2d^2}\right]$$

$$= \frac{\epsilon_0\ a^2}{d}\left[1-\frac{\alpha a}{2d}\right]$$

Q.11 An electron (mass m) with initial velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$ is in an electric field $\vec{E} = -\vec{E}_0\hat{k}$. If λ_0 is initial de-Broglie wavelength of electron, its de-Broglie wave length at time t is given by :

(1) $\frac{\lambda_0\sqrt{2}}{\sqrt{1+\frac{e^2E^2t^2}{m^2v_0^2}}}$ (2) $\frac{\lambda_0}{\sqrt{1+\frac{e^2E^2t^2}{2m^2v_0^2}}}$ (3) $\frac{\lambda_0}{\sqrt{1+\frac{e^2E_0^2t^2}{m^2v_0^2}}}$ (4) $\frac{\lambda_0}{\sqrt{2+\frac{e^2E^2t^2}{m^2v_0^2}}}$

Ans. **[1]**

Sol. $\vec{v} = v_0 \hat{i} + v_0 \hat{j}$

$\vec{E} = -\vec{E}_0 \hat{k}$

$\vec{F} = q\vec{E} = (-e)\ (-E_0 \hat{k}) = eE_0 \hat{k}$

$\therefore\ \vec{a} = \dfrac{\vec{F}}{m} = \dfrac{eE_0}{m}\ \hat{k}$

$\vec{v}$ at time $t = \vec{v} + \vec{a}\,t$

$= v_0 \hat{i} + v_0 \hat{j} + \dfrac{eE_0}{m} t \hat{k}$

$$|\vec{v}| = \sqrt{v_0^2 + v_0^2 + \left(\frac{eE_0 t}{m}\right)^2}$$

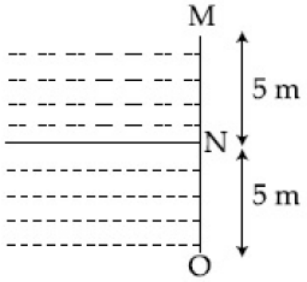
$$= \sqrt{2v_0^2 + \frac{e^2E_0^2}{m^2}t^2}$$

$$\lambda = \frac{h}{mv} = \frac{h}{m\sqrt{2v_0^2 + \frac{e^2E_0^2}{m^2}t^2}}$$

$$= \frac{h}{\sqrt{2}mv_0\sqrt{1 + \frac{e^2E_0^2}{2m^2v_0^2}t^2}}$$

$$= \frac{\lambda_0}{\sqrt{1 + \frac{e^2E_0^2}{2m^2V_0^2}t^2}}$$

Q.12 Two liquids of densities ρ_1 and ρ_2 ($\rho_2 = 2\rho_1$) are filled up behind a square wall of side 10 m as shown in figure. Each liquid has a height of 5m. The ratio of the forces due to these liquids exerted on upper part MN to that at the lower part NO is (Assume that the liquids are not mixing) :



(1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{2}{3}$ (4) $\frac{1}{4}$

Ans. **[4]**

Sol. $F_{MN} = \dfrac{\rho_1 gh}{2} \times a \times h$

$$F_{NO} = \left(\rho_1 gh + \frac{\rho_2 gh}{2}\right) \times a \times h$$

$$\frac{F_{MN}}{F_{ND}} = \frac{\frac{\rho_1}{2}}{\rho_1 + \frac{\rho_2}{2}} = \frac{\rho_1}{2\rho_1 + \rho_2}$$

$$= \frac{\rho_1}{2\rho_1 + 2\rho_1}$$

$$= \frac{1}{4}$$

Q.13 A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the pendulum is 25.0 cm and a stop watch with 1 resolution measures the time taken for 40 oscillations to be 50 s. The accuracy in g is :

(1) 5.40 % (2) 2.40 %

(3) 3.40 % (4) 4.40 %

Ans. **[4] [NTA Answer key (1) 5.40 %]**

Sol. $T = 2\pi\sqrt{\frac{L}{g}}$

$$\Rightarrow T^2 = 4\pi^2 \frac{L}{g}$$

$$\Rightarrow g = 4\pi^2 \frac{L}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + \frac{2\Delta T}{T}$$

$$= \frac{0.1}{25.0} + 2 \times \frac{1}{50}$$

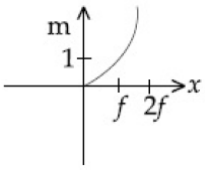
$$= \frac{1}{250} + \frac{1}{25}$$

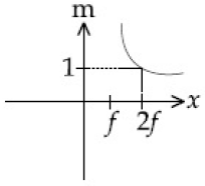
$$= \frac{1+10}{250} = \frac{11}{250}$$

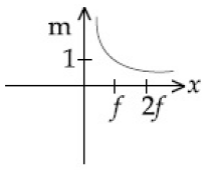
$$\therefore \text{ \% age accuracy} = \frac{11}{250} \times 100\ \%$$

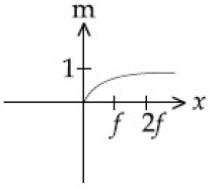
$$= 4.40\ \%$$

Q.14 An object is gradually moving away from the focal point of a concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (m) versus distance of the object from the mirror (x) is correctly given by (Graphs are drawn schematically and are not to scale)

(1) 

(2) 

(3) 

(4) 

Ans. **[2]**

Sol. $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$\Rightarrow \frac{1}{v}=\frac{1}{f}-\frac{1}{u}$

$=\frac{u-f}{uf}$

$\Rightarrow v=\frac{uf}{u-f}$

$m=\frac{-v}{u}=\frac{-f}{u-f}=\frac{f}{f-u}$

as u approaches f, m approaches infinity

at $u = 2f$, $m = -1$

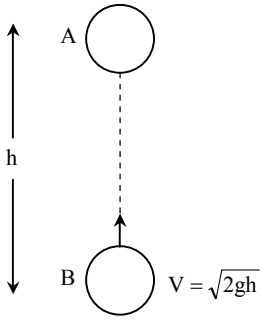
$\Rightarrow |m| = 1$

Q.15 A particle of mass m is dropped from a height a above the ground. At the same time another particle of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{2gh}$. If they collide head-on completely ineleastically, the time taken for the combined mass to reach the ground, in units of $\sqrt{\frac{h}{g}}$ is :

(1) $\frac{1}{2}$ (2) $\sqrt{\frac{3}{2}}$ (3) $\sqrt{\frac{1}{2}}$ (4) $\sqrt{\frac{3}{4}}$

Ans. **[2]**

Sol. Collision takes place at time $= t_1 = \frac{h}{\sqrt{2gh}}$



At $t = t_1$

$V_A = u_A + a_A t_1$

$= 0 + g \times \frac{h}{\sqrt{2gh}}$

$= \sqrt{\frac{gh}{2}}$

$V_B = \sqrt{2gh} - g \times \frac{h}{\sqrt{2gh}}$

$= \sqrt{2gh} - \sqrt{\frac{gh}{2}}$

$= \sqrt{gh}\left[\sqrt{2} - \frac{1}{\sqrt{2}}\right]$

$= \sqrt{\frac{gh}{2}}$

conserving linear momentum just before and just after collision

$m\sqrt{\frac{gh}{2}} - m\sqrt{\frac{gh}{2}} = 2mV$

[V = 0]

height from ground where collision takes place = $h_1 = h - \frac{1}{2}gt_1^2$

$= h - \frac{1}{2}g \cdot \frac{h}{2g} = \frac{3h}{4}$

time taken = $\sqrt{\frac{2h_1}{g}} = \sqrt{\frac{2}{g}.\frac{3h}{4}} = \sqrt{\frac{3h}{2g}}$

Q.16 A transverse wave travels on a taut steel wire with a velocity of v when tension in it is 2.06×10^4 N. When the tension is changed to T, the velocity changed to v/2. The value of T is closed to :

(1) 30.5×10^4 N (2) 5.15×10^3 N

(3) 2.50×10^4 N (4) 10.2×10^2 N

Ans. **[2]**

Sol. $V \propto \sqrt{\text{Tension}}$

$\Rightarrow \frac{V}{V/2} = \sqrt{\frac{2.06\times10^4}{T}}$

$\Rightarrow 2 = \sqrt{\frac{2.06\times10^4}{T}}$

$\Rightarrow 4 = \frac{2.06\times10^4}{T}$

$\Rightarrow T = \frac{2.06}{4}\times10^4$

$\Rightarrow T = \frac{20.6}{4}\times10^3$

$= 5.15 \times 10^3$ N

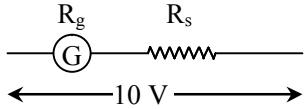
Q.17 A galvanometer having a coil resistance 100 Ω gives a full scale deflection when a current of 1 mA is passed through it. What is the value of the resistance which can convert this galvanometer into a voltmeter giving full scale deflection for a potential difference of 10 V ?

(1) 8.9 kΩ (2) 7.9 kΩ

(3) 9.9 kΩ (4) 10 kΩ

Ans. **[3]**

Sol.

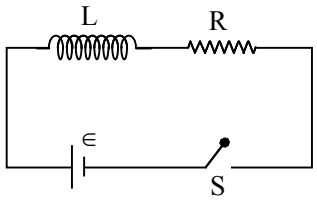


$10 = (1mA) \times (100 + R_s)$

$\Rightarrow 10,000 = 100 + R_s$

$\Rightarrow R_s = 9.9$ kΩ

Q.18 As shown in the figure, a battery of emf $\in$ is connected to an inductor L and resistance R in series. The switch is closed at t = 0. The total charge that flows from battery between t = 0 and t = t_C (t_C is the time constant of the circuit) is :



(1) $\frac{\in R}{eL^2}$ (2) $\frac{\in L}{eR^2}$ (3) $\frac{\in L}{R^2}$ (4) $\frac{\in L}{R^2}\left(1-\frac{1}{e}\right)$

Ans. **[2]**

Sol. $i=\frac{\in}{R}(1-e^{-t/t_c})$

$$\int_0^Q dq=\int_0^{t_c}\frac{\in}{R}(1-e^{-t/t_c})\,dt$$

$$Q=\frac{\in}{R}\left[t-\frac{e^{-t/t_c}}{\left(\frac{-1}{t_c}\right)}\right]_0^{t_c}$$

$$\frac{\in}{R}[(t_c+t_ce^{-1})-(0+t_c)]$$

$$=\frac{\in}{R}t_c.\frac{1}{e}$$

$$=\frac{\in}{R}.\frac{L}{R}.\frac{1}{e}$$

$$=\frac{\in L}{eR^2}$$

Q.19 In a double-slit experiment, at a certain point on the screen the path difference between the two interfering waves is $\frac{1}{8}$th of a wavelength. The ratio of the intensity of light at that point to that at the centre of a bright fringe is :

(1) 0.568 (2) 0.853 (3) 0.672 (4) 0.760

Ans. **[2]**

Sol. $\Delta X=\frac{\lambda}{8}$

$$\Delta\phi=\frac{2\pi}{\lambda}\Delta X=\frac{2\pi}{\lambda}.\frac{\lambda}{8}=\frac{\pi}{4}$$

$$I=4I_0\cos^2\left(\frac{\Delta\phi}{2}\right)$$

$$=4I_0\cos^2\left(\frac{\pi}{8}\right)$$

$$\frac{I}{I_{centre}}=\cos^2\left(\frac{\pi}{8}\right)=0.853$$

Q.20 A plane electromagnetic wave of frequency 25 GHz is propagating in vacuum along the z-direction. At a particular point in space and time, the magnetic field is given by $\vec{B} = 5 \times 10^{-8}\ \hat{j}$ T. The corresponding electric field $\vec{E}$ is (speed of light c = 3×10^8 ms^{-1})

(1) $1.66 \times 10^{-16}\ \hat{i}$ V/m (2) $15\ \hat{i}$ V/m

(3) $-1.66 \times 10^{-16}\ \hat{i}$ V/m (4) $-15\ \hat{i}$ V/m

Ans. **[2]**

Sol. $\vec{B} = 5 \times 10^{-8}\hat{j}$

wave propagates in the direction of $\vec{E} \times \vec{B}$

$\therefore$ $\vec{E}$ is in positive X-direction

$E = cB$

$= 3 \times 10^8 \times 5 \times 10^{-8}$

$= 15$

Q.21 Three containers C_1, C_2 and C_3 have water at different temperatures. The table below shows the final temperature T when different amounts of water (given in liters) are taken from each container and mixed (assume no loss of heat during the process)

C_1	C_2	C_3	T
1ℓ	2ℓ	– –	60°C
– –	1ℓ	2ℓ	30°C
2ℓ	– –	1ℓ	60°
1ℓ	1ℓ	1ℓ	θ

The value of θ (in °C to the nearest integer) is ____ _____

Ans. **[50.00]**

Sol. Let containers C_1, C_2, C_3 contain water at θ_1, θ_2 and θ_3 temperatures respectively.

(i) $1 \times (\theta_1 - 60) + 2(\theta_2 - 60) = 0$

$\Rightarrow \theta_1 + 2\theta_2 = 180$ .....(1)

(ii) $1(\theta_2 - 30) + 2(\theta_3 - 30) = 0$

$\Rightarrow \theta_2 + 2\theta_3 = 90$ .....(2)

(iii) $2(\theta_1 - 60) + 1 \times (\theta_3 - 60) = 0$

$\Rightarrow 2\theta_1 + \theta_3 = 180$ .....(3)

from (1), (2), (3)

$\theta_1 = 80°C$

$\theta_2 = 50°C$

$\theta_3 = 20°C$

(iv) $1 \times (\theta_1 - \theta) + 1 \times (\theta_2 - \theta) + 1(\theta_3 - \theta) = 0$

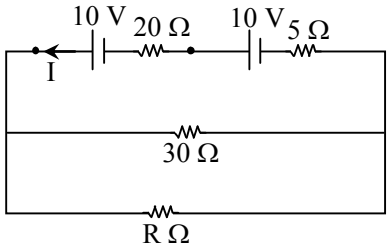
$\theta_1 + \theta_2 + \theta_3 = 30$

$\theta = \frac{\theta_1 + \theta_2 + \theta_3}{3} = 50°C$

Q.22 The series combination of two batteries, both of the same emf 10 V, but different internal resistance of 20 Ω and 5 Ω, is connected to the parallel combination of two resistors 30 Ω and R Ω. The voltage difference across the battery of internal resistance 20 Ω is zero, the value of R (in Ω) is ___ _____

Ans. **[30.00]**

Sol. :



$10 - I \times 20 = 0 \Rightarrow I = 0.5A$

also,

$$I = \frac{20}{20 + 5 + \frac{30R}{30 + R}} = \frac{1}{2}$$

$$\Rightarrow 40 = 25 + \frac{30R}{30 + R}$$

$$\Rightarrow 15 = \frac{30R}{30 + R}$$

$\Rightarrow 30 + R = 2R$

$\Rightarrow [R = 30\Omega]$

Q.23 A ball is dropped from the top of a 100 m high tower on a planet. In the last $\frac{1}{2}$ s before hitting the ground, it covers a distance of 19 m. Acceleration due to gravity (in ms^{-2}) near the surface on that planet is ___ _____.

Ans. **[8.00]**

Sol. Let, the time taken by ball to hit the ground be 't'

$$h = 100 = \frac{1}{2}gt^2 \qquad \text{.....(1)}$$

$$19 = \frac{1}{2}gt^2 - \frac{1}{2}g\left(t - \frac{1}{2}\right)^2$$

$$= \frac{1}{2}g\left(2t - \frac{1}{2}\right) \times \frac{1}{2} \qquad \text{.....(2)}$$

from (2) $152 = g(4t - 1)$

$$\Rightarrow t = \frac{1}{4}\left(1 + \frac{152}{g}\right)$$

putting into (1)

$$100 = \frac{1}{2}g\,\frac{1}{16}\left(1 + \frac{152}{g}\right)^2$$

solving this equation, we get

$g = 8 m/s^2$

Q.24 An asteroid is moving directly towards the centre of the earth. When at a distance of 10 R (R is the radius of the earth) from the earths centre, it has a speed of 12 km/s. Neglecting the effect of earths atmosphere, what will be the speed of the asteroid when it hits the surface of the earth (escape velocity from the earth is 11.2 km/s) ? Give your answer to the nearest integer in kilometer /s ___ _____ _____

Ans. **[16.00]**

Sol. $U_1 + K_1 = U_2 + K_2$

$$\Rightarrow \frac{-GMm}{10R} + \frac{1}{2}mV_1^2 = \frac{-GMm}{R} + \frac{1}{2}mV_2^2$$

$$\Rightarrow \frac{1}{2}V_2^2 = \frac{1}{2}V_1^2 + \frac{GM}{R} - \frac{GM}{10R}$$

$$\Rightarrow V_2^2 = V_1^2 + \frac{9}{5}\frac{GM}{R}$$

$$= 144 + \frac{9}{5}\frac{GM}{R} \quad \text{... ...(1)}$$

Given: $V_e = 11.2$ km/s

$$\Rightarrow \sqrt{\frac{2GM}{R}} = 11.2$$

$$\Rightarrow \frac{GM}{R} = \frac{(11.2)^2}{2} \quad \text{.....(2)}$$

from (1) & (2)

$V_2 = 16.028$ km/s

≈ 16 km/s

Q.25 The first member of the Balmer series of hydrogen atom has a wavelength of the 6561 Å. The wavelength of the second member of the Balmer series (in nm) is _____.

Ans. **[486.00]**

Sol. $$\frac{1}{\lambda_1} = RZ^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5}{36}RZ^2$$

$$\frac{1}{\lambda_2} = RZ^2\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = \frac{12}{64}RZ^2$$

$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{5}{36} \times \frac{64}{12} = \frac{20}{27}$$

$$\Rightarrow \lambda_2 = \frac{20}{27}\lambda_1$$

$$= \frac{20}{27} \times 6561 \text{ Å}$$

$= 486$ nm

JEE Main Online Exam 2020

Questions & Solution

8th January 2020 | Shift - II

CHEMISTRY

Q.1 The increasing order of the atomic radii of the following elements is :

(a) C (b) O (c) F (d) Cl

(e) Br

(1) (a) < (b) < (c) < (d) < (e) (2) (d) < (c) < (b) < (a) < (e)

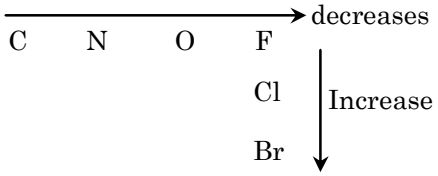
(3) (c) < (b) < (a) < (d) < (e) (4) (b) < (c) < (d) < (a) < (e)

Ans. **[3]**

Sol. Generally in a period L → R Atomic radius decrease

and

in a Group T → B Atomic radius increase



Br > Cl > C > O > F

e > d > a > b > c

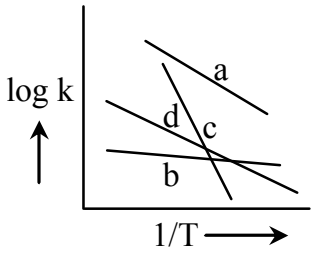
Q.2 Kjeldahl's method cannot be used to estimate nitrogen for which of the following compounds?

(1) $NH_2 - \overset{\overset{O}{\|}}{C} - NH_2$ (2) $C_6H_5NH_2$ (3) $C_6H_5NO_2$ (4) $CH_3CH_2 - C \equiv N$

Ans. **[3]**

Sol. Kjedahl's method cannot be used to Test nitrogen in nitro and diazo present in ring because nitrogen in nitro cannot convert into Ammonium sulphate Hence. does not give Test.

Q.3 Consider the following plots of rate constant versus $\frac{1}{T}$ for four different reactions. which of the following orders is correct for the activation energies of these reactions?



(1) $E_c > E_a > E_d > E_b$ (2) $E_b > E_d > E_c > E_a$ (3) $E_a > E_c > E_d > E_b$ (4) $E_b > E_a > E_d > E_c$

Ans. **[1]**

Sol. We know that

$k = Ae^{-\frac{Ea}{RT}}$

$\log^{k} = \log^{A} - \frac{Ea}{2.303RT}$

$\text{Slope} = -\frac{Ea}{2.303RT}$

Slope $c > a > d > b$.

$E_c > E_a > E_d > E_b$

Q.4 For the following Assertion and Reason, the correct option is :

Assertion : The pH of water increase with increase in temperature.

Reason : The dissociation of water into H^+ and OH^- is an exothermic reaction

(1) Assertion is not true, but reason is true

(2) Both assertion and reason are false

(3) but the reason is not the correct explanation for the assertion

(4) Both assertion and reason are true, and the reason is the correct explanation for the assertion

Ans. **[2]**

Sol. $kw = [H^+]\,[OH^-]$

$T \uparrow\ kw \uparrow$

In water $[H^+] = [OH^-]$

$kw = [H^+]^2 \Rightarrow [H^+] = \sqrt{kw}$

On Increasing T kw $\uparrow$ $[H^+] \uparrow$ $P^H \downarrow$ Assertion is False.

Dissociation of H_2O is endothermic.

$H_2O \rightleftharpoons H^+ + OH^-$ $\Delta H = +\text{ ve}$

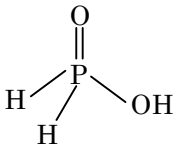
Hence Reason is also false.

Q.5 White phosphorus on reaction with concentrated NaOH solution in an inert atmosphere of CO_2 gives phosphine and compound (X). (X) on acidification with HCl gives compound (Y). The basicity of compound (Y) is :

(1) 2 (2) 4 (3) 3 (4) 1

Ans. **[4]**

Sol. $P_4 + NaOH + H_2O \longrightarrow PH_3 + NaH_2PO_2 \xrightarrow{HCl} H_3PO_2$



Basicity = 1

Q.6 The major product [B] in the following sequence is :

$CH_3-\underset{\underset{CH(CH_3)_2}{|}}{C}=CH-CH_2CH_3 \xrightarrow[(ii)\ H_2O_2,\ O\overset{\ominus}{H}]{(i)\ B_2H_6} [A] \xrightarrow[\Delta]{dil.\ H_2SO_4} [B]$

(1) $CH_3-\underset{\underset{CH(CH_3)_2}{|}}{CH}-CH=CH-CH_3$

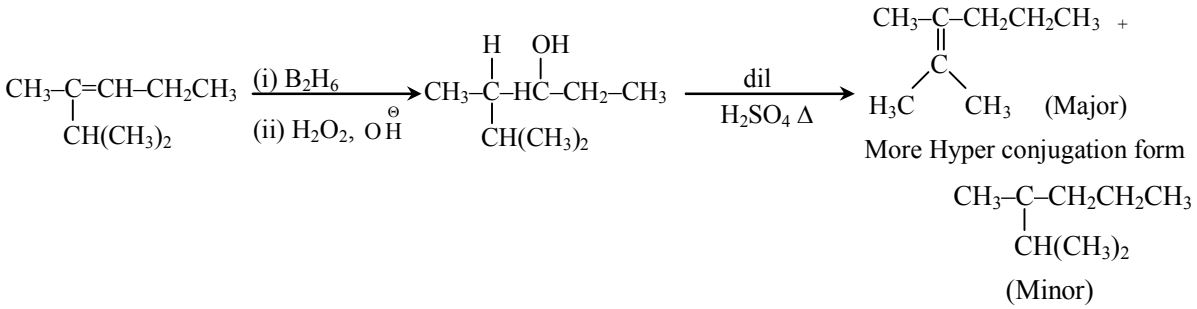
(2) $CH_3-\underset{\underset{CH(CH_3)_2}{|}}{C}=CH-CH_2CH_3$

(3) $CH_2=\underset{\underset{CH(CH_3)_2}{|}}{C}-CH_2CH_2CH_3$

(4) $CH_3-\underset{\underset{C}{\|}}{C}-CH_2CH_2CH_3$ with C bonded to H_3C and CH_3

Ans. **[4]**

Sol.



More Hyper conjugation form

(Minor)

Q.7 Among (a) – (d), the complexes that can display geometrical isomerism are :

(a) $[Pt(NH_3)_3Cl]^+$ (b) $[Pt(NH_3)Cl_5]^-$ (c) $[Pt(NH_3)_2Cl(NO_2)]$ (d) $[Pt(NH_3)_4ClBr]^{2+}$

(1) (d) and (a) (2) (c) and (d) (3) (b) and (c) (4) (a) and (b)

Ans. **[2]** (c) and (d)

Sol. $[Pt(NH_3)_2Cl(NO_2)]$ & $[Pt(NH_3)_4ClBr]^{2+}$

$[M A_2 BC]$ type

(AA) (BC)

(AB) (AC)

2 G.I.

$[M A_4 BC]$ type

(AA) (AA) (BC)

(AB) (AC) (AA)

2 G.I.

Hence these two show G.I.

Q.8 For the following Assertion and Reason, the correct option is :

Assertion : For hydrogenation reactions, the catalytic activity increases from Group 5 to Group 11 metals with maximum activity shown by Group 7-9 elements.

Reason : The reactants are most strongly adsorbed on group 7-9 elements.

(1) Both assertion and reason are true and the reason is the correct explanation for the assertion.

(2) The assertion is true, but the reason in false.

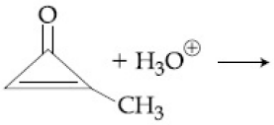
(3) Both assertion and reason are true but the reason is not the correct explanation for the assertion.

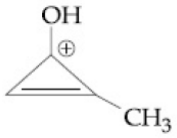
(4) Both assertion and reason are false.

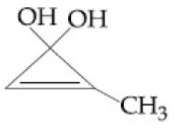
Ans. **[1]**

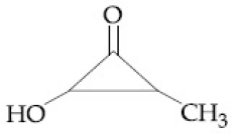
Sol. For hydrogenation reaction catalytic activity increase because reactants are more strongly adsorbed on group 7-9 element, So Assertion & Reason both are correct.

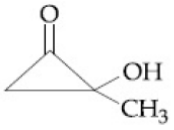
Q.9 The major product in the following reaction is :



(1) 

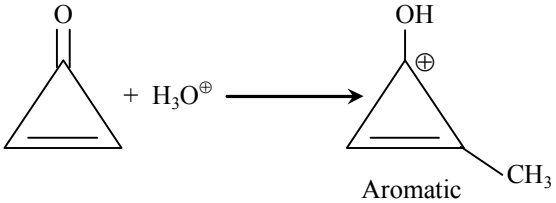
(2) 

(3) 

(4) 

Ans. **[1]**

Sol.



Aromatic

Q.10 Among the reactions (a)-(d), the reaction(s) that does/do not occur in the blast furnace during the extraction of iron is/are :

(a) $CaO + SiO_2 \rightarrow CaSiO_3$ (b) $3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$

(c) $FeO + SiO_2 \rightarrow FeSiO_3$ (d) $FeO \rightarrow Fe + \frac{1}{2}O_2$

(1) (c) and (d) (2) (a) (3) (a) and (d) (4) (d)

Ans. **[1]**

Sol. $CaO + SiO_2 \rightarrow CaSiO_3$

Used as flux.

$3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$

Reduction done by CO.

Hence these two r × n take place but

$FeO \rightarrow Fe + \frac{1}{2}O_2$ & $FeO + SiO_2 \rightarrow FeSiO_3$

does not take place.

Q.11 A metal (A) on heating in nitrogen gas gives compound B. B on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a dark blue-violet coloured solution. A and B respectively, are :

(1) Mg and $Mg(NO_3)_2$ (2) Na and $NaNO_3$ (3) Mg and Mg_3N_2 (4) Na and Na_3N

Ans. **[3]**

Sol. $A + N_2 \longrightarrow$ nitride (B) $\xrightarrow{H_2O} NH_3 + CuSO_4 \longrightarrow$ Blue-violet coloured sol.

$\underset{(A)}{3Mg} + N_2 \longrightarrow \underset{(B)}{Mg_3N_2} \xrightarrow{H_2O} Mg(OH)_2 + NH_3$

Q.12 Two monomers in maltose are :

(1) α-D-glucose and α-D-galactose (2) α-D-glucose and β-D-glucose

(3) α-D-glucose and α-D-glucose (4) α-D-glucose and α-D-Fructose

Ans. **[3]**

Sol. Maltose on hydrolysis give 2 mole of α-D Glucose.

because in maltose glucosidic linkage is present in between C_1 & C_4 of α-D glucose.

Q.13 The correct order of the calculated spin-only magnetic moments of complexes (A) to (D) is :

(A) $Ni(CO)_4$ (B) $[Ni(H_2O)_6]Cl_2$ (C) $Na_2[Ni(CN)_4]$ (4) $PdCl_2(PPh_3)_2$

(1) (C) ≈ (D) < (B) < (A) (2) (A) ≈ (C) < (B) ≈ (D)

(3) (A) ≈ (C) ≈ (D) < (B) (4) (C) < (D) < (B) < (A)

Ans. **[3]**

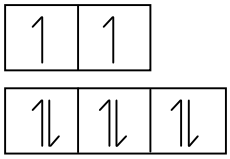
Sol. (a) $Ni(CO)_4$

$Ni = 3d^8\ 4s^2$ [S F L] M = 0

(b) $[Ni(H_2O)_6]\ Cl_2$

$Ni^{2+} = 3d^8\ 45°$

Weak field ligand



No. of unpaired electron = 2

$M = \sqrt{n(n+2)} = \sqrt{2(2+2)} = \sqrt{8}$ B.M

(c) $Na_2[Ni(CN)_4]$

$Ni^{2+} - 3d^8$ (S F L)

paining take place (n = 0)

M = 0

(d) $Pd\,Cl_2(PPh_3)_2$

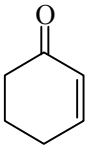
$Pd^{2+} = 4d^8$

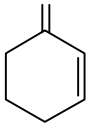
square plane

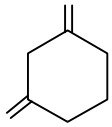
M = 0

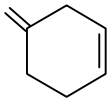
Q.14 An unsaturated hydrocarbon X absorbs two hydrogen molecules on catalytic hydrogenation, and also gives following reaction :

$X \xrightarrow[Zn/H_2O]{O_3} A \xrightarrow{[Ag(NH_3)_2]^+} B$(3-oxo-hexanedicarboxylic acid) X will be :

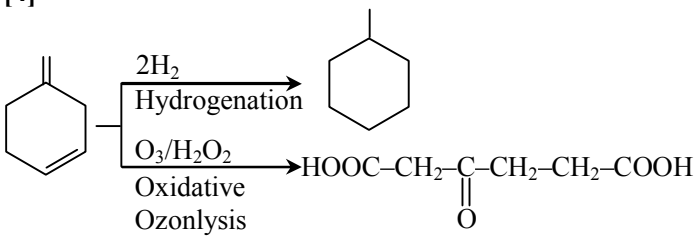
(1) 

(2) 

(3) 

(4) 

Ans. **[4]**

Sol. 

Q.15 Hydrogen has three isotopes (A), (B) and (C). If the number of neutron(s) in (A), (B) and (C) respectively, are (x), (y) and (z), the sum of (x), (y) and (z) is :

(1) 4 (2) 3 (3) 2 (4) 1

Ans. **[2]**

Sol.

	^1_1H	^2_1H	^3_1H
no. of neutron	0 (x)	1 (y)	1 (z)

= x + y + z

= 0 = 1 + 2 = 3

Q.16 Which of the following compound is likely to show both Frenkel and Schottky defects in its crystalline form?

(1) CsCl (2) AgBr (3) ZnS (4) KBr

Ans. **[2]**

Sol. AgBr show both Frenkel and Schottky defect.

Q.17 Arrange the following bonds according to their average bond energies in descending order:

C–Cl, C–Br, C–F, C–I

(1) C–F > C–Cl > C–Br > C–I (2) C–I > C–Br > C–Cl > C–F

(3) C–Cl > C–Br > C–I > C–F (4) C–Br > C–I > C–Cl > C–F

Ans. **[1]**

Sol. Bond energy $\propto \frac{1}{\text{Bond length}}$.

Down the group size increases

Bond energy C–F > C–Cl > C–Br > C–I

Q.18 Preparation of Bakelite proceeds via reactions :

(1) Electrophilic substitution and dehydration (2) Condensation and elimination

(3) Electrophilic addition and dehydration (4) Nucleophilic addition and dehydration

Ans. **[1]**

Sol. Formation of Bakelite follows electrophillic substitution of phenol and formaldehyde followed by dehydration .

Q.19 The radius of the second Bohr orbit, in terms of the Bohr radius, a_0, in Li^{2+} is :

(1) $\frac{4a_0}{3}$ (2) $\frac{2a_0}{9}$ (3) $\frac{2a_0}{3}$ (4) $\frac{4a_0}{9}$

Ans. **[1]**

Sol. We know

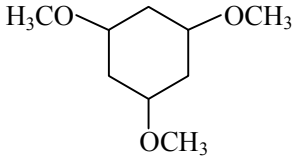
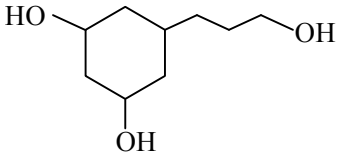
$$r = 0.529 \times \frac{n^2}{z} Å$$

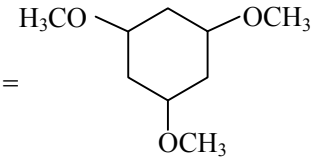
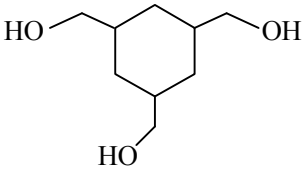
$$r = a_0 \times \frac{n^2}{z}$$

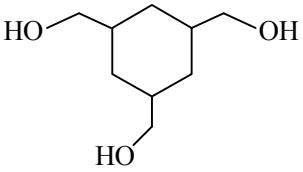
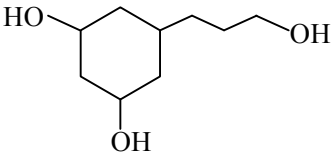
$n = 2$ $z = 3$

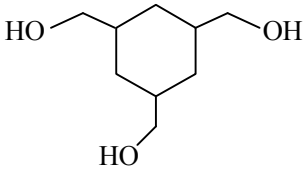
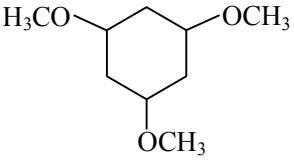
$r = r = \frac{a_0 \times 4}{3} = \frac{4a_0}{3}$

Q.20 Among the compounds A and B with molecular formula $C_9H_{18}O_3$, A is having higher boiling point the B. The possible structures of A and B are :

(1) A =  ; B = 

(2) A =  ; B = 

(3) A =  ; B = 

(4) A =  ; B = 

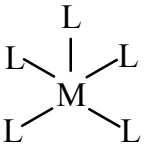
Ans. **[4]**

Sol. A is having higher boiling point than B. in case of A inter molecular H-bonding is possible while in case of B. intermolecular H-bonding is not possible hence have lower boiling point

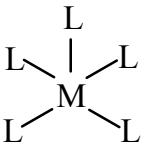
Q.21 Complexes (ML_5) of metals Niand Fe have ideal square pyramidal and trigonal bipyramidal geometries, respectively. The sum of the 90°, 120° and 180° L-M-L angles in the two complexes is _____

Ans. **[20.00]**

Sol.



90° = 8
180° = 8
total = 10



12° = 3
90° = 6
total = 1

Q.22 At constant volume, 4 mol of an ideal gas when heated from 300 K to 500 K changes its internal energy by 5000 J. The molar heat capacity at constant volume is _____

Ans. **[6.25]**

Sol. $\Delta U = ncv\,\Delta T$

$5000 = 4 \times C_v\,(500 - 300)$

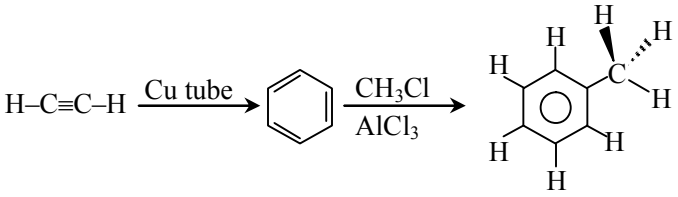
$C_v = 6.25\ J\ k^{-1}\ mol^{-1}$

Q.23 In the following sequence of reactions the maximum number of atoms present in molecule 'C' in one plane is _____.

$$A \xrightarrow[\text{Cu tube}]{\text{Red hot}} B \xrightarrow[\text{Anhydrous } AlCl_3]{CH_3Cl(1.eq.)} C$$ (A is a lowest molecular weight alkyne)

Ans. **[13.00]**

Sol.



Q.24 For an electrochemical cell $Sn(s)|Sn^{2+}$ (aq, 1M)$||Pb^{2+}$ (aq, 1M)$|Pb(s)$ the ratio $\frac{[Sn^{2+}]}{[Pb^{2+}]}$ when this cell attains equilibrium is _____.

$$\left(\text{Given}: E^0_{Sn^{2+}|Sn} = -0.14V, E^0_{Pb^{2+}|Pb} = -0.13V, \frac{2.303RT}{F} = 0.06\right)$$

Ans. **[2.15]**

Sol. At eqll^{m}

$E_{Cell} = 0$

$E^0_{Cell} = 0.01$ V

$Sn + Pb^{2+} \longrightarrow Sn^{2+} + Pb$

$$E_{Cell} = E^0_{Cell} - \frac{0.06}{n} \log^{Q}$$

$$0 = 0.01 - \frac{0.06}{2} \log \frac{[Sn^{2+}]}{[Pb^{2+}]}$$

$$0.01 = \frac{0.06}{2} \log \frac{[Sn^{2+}]}{[Pb^{2+}]}$$

$$\frac{1}{3} = \log \frac{[Sn^{2+}]}{[Pb^{2+}]}$$

$$\frac{[Sn^{2+}]}{[Pb^{2+}]} = 10^{1/3} = 2.15$$

Q.25 $NaClO_3$ is used, even in spacecraft, to produce O_2. The daily consumption of pure O_2 by a person is 492L at 1 atm, 300 K. How much amount of $NaClO_3$, in grams, is required to produce O_2 for the daily consumption of a person at 1 atm, 300 K? _____.

$NaClO_3(s) + Fe(s) \rightarrow O_2(g) + NaCl(s) + FeO(s)$

$R = 0.082$ L atm mol^{-1} K^{-1}

Ans. **[2130.00]**

Sol. Mole of $NaCl_3$ = mole of O_2

$$\text{mole of } O_2 = \frac{PV}{RT} = \frac{1 \times 492}{0.082 \times 300} = 20 \text{ mole}$$

mass of $NaClO_3 = 20 \times 106.5 = 2130$ g

JEE Main Online Exam 2020

Questions & Solutions

8th January 2020 | Shift - II

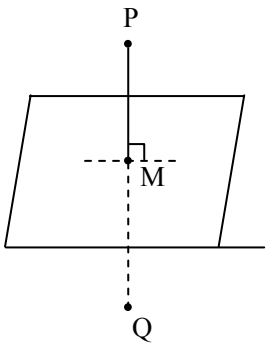
MATHEMATICS

Q.1 The mirror image of the point (1,2,3) in a plane is $\left(-\frac{7}{3},-\frac{4}{3},-\frac{1}{3}\right)$.Which of the following points lies on this plane ?

(1) (1, – 1, 1) (2) (–1, – 1, 1) (3) (1,1,1) (4) (–1, –1, –1)

Ans. **[1]**

Sol.



$$\text{Mid point M} = \left(\frac{-2}{3},\frac{2-\frac{4}{3}}{2},\frac{3-\frac{1}{3}}{2}\right)$$

$$= \left(\frac{-2}{3},\frac{1}{3},\frac{4}{3}\right)$$

$$\text{DR of PM} = \left(1+\frac{2}{3},2-\frac{1}{3},3-\frac{4}{3}\right)$$

$$= \left(\frac{5}{3},\frac{5}{3},\frac{5}{3}\right)$$

$$= (1,1,1)$$

$$\text{Normal} = (\bar{r}-\bar{a}).\ \bar{n} = 0$$

$$\left(x+\frac{2}{3}\right)+\left(y-\frac{1}{3}\right)+\left(z-\frac{4}{3}\right) = 0$$

$x + y + z = 1$

(1, –1, 1) satisfies this plane

Q.2 If α and β be the coefficients of x^4 and x^2 respectively in the expansion of $\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6$, then :

(1) $\alpha - \beta = -132$ (2) $\alpha + \beta = -30$ (3) $\alpha + \beta = 60$ (4) $\alpha - \beta = 60$

Ans. **[1]**

Sol. $2[{}^6C_0\, x^6 + {}^6C_2\, x^4\,(x^2-1) + {}^6C_4\, x^2\,(x^2-1)^2 + {}^6C_6\,(x^2-1)^3]$

$\alpha = 2\,[-{}^6C_2 - 2\,{}^6C_4 - 3] = -2\,[15 + 30 + 3] = -96$

$\beta = 2\,[{}^6C_4 + 3] = 2[15 + 3] = 36$

$\alpha - \beta = -96 - 36 = -132$

Q.3 The area (in sq. units) of the region $\{(x, y) \in R^2 : x^2 \le y \le 3 - 2x\}$, is :

(1) $\frac{32}{3}$ (2) $\frac{29}{3}$ (3) $\frac{31}{3}$ (4) $\frac{34}{3}$

Ans. **[1]**

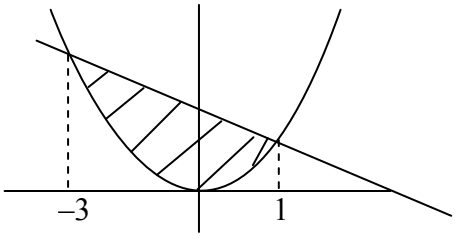
Sol. $x^2 \le y \le -2x + 3$

$x^2 = -2x + 3$

$x^2 + 2x - 3 = 0$

$(x + 3)(x - 1) = 0$

$x = -3, x = 1$



Area $= \int_{-3}^{1} (-2x + 3 - x^2)dx$

$= \left(-x^2 + 3x - \frac{x^3}{3}\right)_{-3}^{1}$

$= \left(-1 + 3 - \frac{1}{3} + 9 + 9 - 9\right)$

$= 11 - \frac{1}{3} = \frac{32}{3}$ sq. units

Q.4 Let $f : (1, 3) \to R$ be a function defined by $f(x) = \frac{x[x]}{1+x^2}$, where [x] denotes the greatest integer $\le x$. Then the range of f is :

(1) $\left(\frac{2}{5}, \frac{3}{5}\right] \cup \left(\frac{3}{4}, \frac{4}{5}\right)$ (2) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left[\frac{3}{5}, \frac{4}{5}\right]$

(3) $\left(\frac{2}{5}, \frac{4}{5}\right]$ (4) $\left(\frac{3}{5}, \frac{4}{5}\right)$

Ans. **[2]**

Sol. $f(x) = \frac{x}{x^2+1} = \frac{1}{x+\frac{1}{x}}$ $\quad 1 < x < 2$

$= \frac{2x}{x^2+1} = \frac{2}{x+\frac{1}{x}}$ $\quad 2 \le x < 3$

$= x + \frac{1}{x} \uparrow$ for $x \in (1,\infty)$

So $f(x) \in \left(\frac{2}{5},\frac{1}{2}\right)$; (1, 2)

$\in \ 2 \le x < 3$

$\left(\frac{6}{10},\frac{4}{5}\right) = \left(\frac{3}{5},\frac{4}{5}\right]$

$\left(\frac{2}{5},\frac{1}{2}\right) \cup \left(\frac{3}{5},\frac{4}{5}\right]$

Q.5 Let A and B be two events such that the probability that exactly one of them occurs is $\frac{2}{5}$ and the probability that A or B occurs is $\frac{1}{2}$, then the probability of both of them occur together is :

(1) 0.01 (2) 0.10 (3) 0.02 (4) 0.20

Ans. **[2]**

Sol. $P(A) + P(B) - 2P(A \cap B) = \frac{2}{5}$ … (1)

$P(A) + P(B) - P(A \cap B) = \frac{1}{2}$ … (2)

eq(2) – eq(1)

$P(A \cap B) = \frac{1}{2} - \frac{2}{5} = \frac{5-4}{10} - \frac{1}{10} = 0.10$

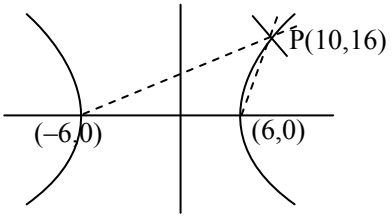
Q.6 If a hyperbola passes through the point P(10, 16) and it has vertices at (± 6, 0), then the equation of the normal to it at P is :

(1) $2x + 5y = 100$ (2) $x + 2y = 42$

(3) $x + 3y = 58$ (4) $3x + 4y = 94$

Ans. **[1]**

Sol.



$a = 6$

$$\frac{100}{36} - \frac{256}{b^2} = 1 \Rightarrow \frac{100}{36} - 1 = \frac{256}{b^2}$$

$$\frac{256}{b^2} = \frac{64}{36}$$

$$b^2 = \frac{256 \times 36}{64}$$

$$\Rightarrow b = \frac{16 \times 6}{8} = 12$$

$$\frac{x^2}{36} - \frac{y^2}{144} = 1$$

eq. of normal $\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$

$$\frac{36x}{10} + \frac{144y}{16} = 36 + 144 = 180$$

$$\frac{4x}{10} + y = 20$$

$2x + 5y = 100$

Q.7 The length of the perpendicular from the origin, on the normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at the point (2, 2) is :

(1) $2\sqrt{2}$ (2) $4\sqrt{2}$ (3) 2 (4) $\sqrt{2}$

Ans. **[1]**

Differentiate the curve

$2x + 2xy' + 2y - 6yy' = 0$

$2 + 2y' + 2 - 6y' = 0$

$y' = +1$

Normal at (2,2)

$y - 2 = -1\,(x - 2)$

$y + x = 4$

distance from 0 is $= \frac{4}{\sqrt{2}} = 2\sqrt{2}$

Q.8 Let $\alpha = \frac{-1 + i\sqrt{3}}{2}$. If $a = (1 + \alpha) \sum_{k=0}^{100} \alpha^{2k}$ and $b = \sum_{k=0}^{100} \alpha^{3k}$, then a and b are the roots of the quadratic equation :

(1) $x^2 + 102x + 101 = 0$ (2) $x^2 - 102x + 101 = 0$

(3) $x^2 - 101x + 100 = 0$ (4) $x^2 + 101x + 100 = 0$

Ans. **[2]**

Sol. $\alpha = \omega = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$

$b = \sum_{k=0}^{100} \alpha^{3k} = \alpha \sum_{k=0}^{100} 1 = 101$

$a = (1 + \alpha) \sum_{k=0}^{100} \alpha^{2k}$

$= (1 + \omega) \frac{1-(\omega^2)^{101}}{1-\omega^2} = \frac{(1+\omega)(1-\omega)}{1-\omega^2} = 1$

roots are 101 & 1

eq = $x^2 - 102x + 101 = 0$

$x^2 - 102\,x + 101 = 0$

Q.9 The system of linear equations

$\lambda x + 2y + 2z = 5$

$2\lambda x + 3y + 5z = 8$

$4x + \lambda y + 6z = 10$ has :

(1) infinitely many solutions when $\lambda = 2$

(2) no solution when $\lambda = 2$

(3) No solution when $\lambda = 8$`

(4) a unique solution when $\lambda = -8$

Ans. **[2]**

Sol. $\Delta = \begin{bmatrix} \lambda & 2 & 2 \\ 2\lambda & 3 & 5 \\ 4 & \lambda & 6 \end{bmatrix}$

$= \lambda (18 - 5\lambda) - 2 (12\lambda - 20) + 2 (2\lambda^2 - 12)$

$= 18\lambda - 5\lambda^2 - 24\lambda + 24\lambda + 40 + 4\lambda^2 - 24$

$= -\lambda^2 - 6\lambda + 16$

If $\Delta = 0 \Rightarrow \lambda^2 + 6\lambda - 10 = 0$

$(\lambda + 8)(\lambda - 2) = 0$

$\lambda = -8$ or $\lambda = 2$

if $\lambda = 2$: $\Delta_x = \begin{bmatrix} 5 & 2 & 2 \\ 8 & 3 & 5 \\ 10 & 2 & 6 \end{bmatrix} = 5 (18 - 10) - 2 (48 - 50) + 2 (16 - 30)$

$= 40 + 4 - 28 \neq 0$

$\Delta_y = \begin{bmatrix} 2 & 5 & 2 \\ 4 & 8 & 5 \\ 4 & 10 & 6 \end{bmatrix}$

$\Delta_z = \begin{bmatrix} 2 & 2 & 5 \\ 4 & 3 & 8 \\ 4 & 2 & 10 \end{bmatrix}$

So no solution for $\lambda = 2$

Q.10 $\lim\limits_{x\to 0}\dfrac{\int_0^x t\sin(10t)dt}{x}$ is equal to :

(1) $-\dfrac{1}{5}$ (2) $\dfrac{1}{10}$ (3) $-\dfrac{1}{10}$ (4) 0

Ans. **[4]**

Sol. differentiate w.r.t. x

$\lim\limits_{x\to 0}\dfrac{x\sin(10x)}{1}=0$

Q.11 Let $\vec{a}=\hat{i}-2\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+\hat{k}$ be two vectors. If $\vec{c}$ is a vector such that $\vec{b}\times\vec{c}=\vec{b}\times\vec{a}$ and $\vec{c}.\vec{a}=0$, then $\vec{c}.\vec{b}$ is equal to :

(1) -1 (2) $\dfrac{1}{2}$ (3) $-\dfrac{1}{2}$ (4) $-\dfrac{3}{2}$

Ans. **[3]**

Sol. $\vec{b}\times(\vec{c}-\vec{a})=0$

$(\vec{c}-\vec{a})=\lambda\vec{b}$

$\vec{c}=\vec{a}+\lambda\vec{b}$

$\vec{a}\cdot\vec{c}=0=6+\lambda 4$

$\lambda=-3/2$

So $\vec{c}=\vec{a}+\dfrac{3}{2}\vec{b}$

$=(\hat{i}-2\hat{j}+\hat{k})-\dfrac{3}{2}(\hat{i}-\hat{j}+\hat{k})$

$2\vec{c}=2\vec{a}-3\vec{b}$

$2\vec{c}=(2\hat{i}-4\hat{j}+2\hat{k})-(3\hat{i}-3\hat{j}+3\hat{k})$

$=-\hat{i}-\hat{j}-\hat{k}$

So $\vec{c}=-\dfrac{1}{2}(\hat{i}+\hat{j}+\hat{k})$

So $\vec{b}\cdot\vec{c}=\dfrac{1}{2}(1-1+1)=-\dfrac{1}{2}$

Q.12 Which of the following statements is a tautology ?

(1) $p\vee(\sim q)\to p\wedge q$ (2) $\sim(p\wedge\sim q)\to p\vee q$

(3) $\sim(p\vee\sim q)\to p\wedge q$ (4) $\sim(p\vee\sim q)\to p\vee q$

Ans. **[4]**

Sol.

p	q	~ p	~ q	p∨ ~ q	~ (p∨ ~ q)	p∨ q	~ (p∨ ~ q) → (p∨ q)
T	T	F	F	T	F	T	T
T	F	F	T	T	F	T	T
F	T	T	F	F	T	T	T
F	F	T	T	T	F	F	T

Q.13 Let S be the set of all functions f : [0,1] → R, which are continuous on [0,1] and differentiable on (0, 1). Then for every f in S, there exists a c ∈ (0, 1), depending on f, such that :

(1) $\frac{f(1)-f(c)}{1-c}=f'(c)$ (2) $|f(c)-f(1)|<(1-c)|f(c)|$

(3) $|f(c)-f(1)|<|f'(c)|$ (4) $|f(c)+f(1)|<(1+c)|f'(c)|$

Ans. **[Bonus]**

Sol. S is set of all functions

If we consider a constant function, then option 2, 3 and 4 are incorrect.

For option 1 :

$\frac{f(1)-f(c)}{1-c}=f'(c)$

This may not be true for all the function

If we apply LMVT in (c, 1) then

∃ c' ∈ (c, 1) such that

$\frac{f(1)-f(c)}{1-c}=f'(c')$

None of the option are incorrect

Q.14 If the 10th term of an A.P is $\frac{1}{20}$ and its 20^{th}term is $\frac{1}{10}$, then the sum of its first 200 terms is :

(1) $50\frac{1}{4}$ (2) $100\frac{1}{2}$ (3) 100 (4) 50

Ans. **[2]**

Sol. $T_{10}=\frac{1}{20}=a+9d$

$T_{20}=\frac{1}{10}=a+19d.$

$\frac{1}{20}-\frac{1}{10}=-10d$

$-10d=\frac{1-2}{20}=-\frac{1}{20}$

$d=\frac{1}{200}$

So $a=\frac{1}{20}-\frac{9}{200}=\frac{10-9}{200}=\frac{1}{200}$

$S=\frac{200}{2}\left[\frac{2}{200}+\frac{199}{200}\right]=100.\frac{201}{200}=\frac{201}{2}$

Q.15 If $A=\begin{pmatrix}2 & 2\\ 9 & 4\end{pmatrix}$ and $I=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$, then $10A^{-1}$ is equal to :

(1) 4I – A (2) 6I – A (3) A – 6I (4) A – 4I

Ans. **[3]**

Sol. A – x I = 0

$\begin{vmatrix}2-x & 2\\ 9 & 4-x\end{vmatrix}=0$

$(2 - x)(4 - x) - 18 = 0$

$8 + x^2 - 6x - 18 = 0$

$x^2 - 6x - 10 = 0$

$[A^2 - 6A - 10I = 0] \times A^{-1}$

$A - 6I - 10A^{-1} = 0$

$10A^{-1} = A - 6I$

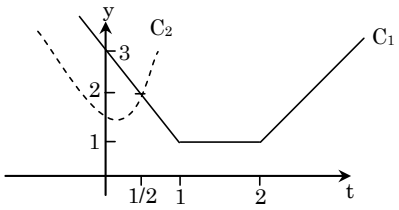
Q.16 Let S be the set of all real roots of the equation, $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S :

(1) is a singleton. (2) contains exactly two elements

(3) is an empty set. (4) contains at least four elements.

Ans. **[1]**

Sol.



C_1: $y = |t - 1| + |t - 2|$

C_2: $y = t^2 - 1 + 2$

Let $3^x = t > 0$

$t(t - 1) + 2 = |t - 1| + |t - 2|$

$t^2 - t + 2 = |t - 1| + |t - 2|$

Minimum value of C_2

$-\frac{D}{4a} = -\frac{1-8}{4} = \frac{7}{4}$

Singleton solution

Q.17 If $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$, then :

(1) $\frac{1}{8} < I^2 < \frac{1}{4}$ (2) $\frac{1}{9} < I^2 < \frac{1}{8}$ (3) $\frac{1}{16} < I^2 < \frac{1}{9}$ (4) $\frac{1}{6} < I^2 < \frac{1}{2}$

Ans. **[2]**

Sol. $\int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$

Let $f(x) = f(x) = \frac{1}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$

$f'(x) = \frac{6x^2 - 18x + 12}{\sqrt{2x^2 - 9x^2 + 12x + 4}}$

$= \frac{6(x - 1)(x - 2)}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$

$+ \quad 1 \quad - \quad 2 \quad +$

$f_{max} = f(1)$

$f_{min} = f(2)$

$(f_1) = \frac{1}{\sqrt{2-9+12+4}} = \frac{1}{\sqrt{9}} = \frac{1}{3}$

$(f_2) = \frac{1}{\sqrt{16-36+24+4}} = \frac{1}{\sqrt{8}}$

$\frac{1}{3} < I < \frac{1}{\sqrt{8}}$

So $\frac{1}{9} < I^2 < \frac{1}{8}$

Q.18 The differential equation of the family of curves, $x^2 = 4b(y + b)$, $b \in R$, is :

(1) $x(y')^2 = 2yy' - x$ (2) $x(y')^2 = x - 2yy'$

(3) $x(y')^2 = x + 2yy'$ (4) $xy'' = y'$

Ans. **[3]**

Sol. $\mathbf{x^2} = 4b\,(y + b)$

$2x = 4by' \Rightarrow b = \frac{x}{2y'}$

$2x = \frac{4x}{2y'}\left(y + \frac{x}{2y'}\right)$

$x = \frac{2}{2(y')^2}(2yy' + x)$

$x\,(y')^2 = 2yy' + x$

Q.19 If a line, $y = mx + c$ is a tangent to the circle, $(x - 3)^2 + y^2 = 1$ and it is perpendicular to a line L_1, where L_1 is the tangent to the circle, $x^2 + y^2 = 1$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$; then :

(1) $c^2 + 7c + 6 = 0$ (2) $c^2 + 6c + 7 = 0$ (3) $c^2 - 6c + 7 = 0$ (4) $c^2 - 7c + 6 = 0$

Ans. **[2]**

Sol. tangent to c_2 is

$\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} = 1$

Slope $= -1$

So here $m = 1$

So tangent to c_1 is $y = x + c$

distance from (3, 0) is

$\frac{|3 + c|}{\sqrt{1+1}} = 1$

$9 + c^2 + 6c = 2$

$c^2 + 6c + 7 = 0$

Q.20 The mean and variance of 20 observations are found to be 10 and 4, respectively. On rechecking, it was found that an observation 9 was incorrect and the correct observation was 11. Then the correct variance is :

(1) 3.99 (2) 3.98 (3) 4.01 (4) 4.02

Ans. **[1]**

Sol. $\frac{\sum x_i}{20} = 10$

$\sum x_i = 20$

$\frac{\sum x_i^2}{20} - 100 = 4$

$\sum x_i^2 = 2080$

actual mean $\frac{200-9+11}{20} = \frac{202}{20} = \frac{101}{10} = 10.1$

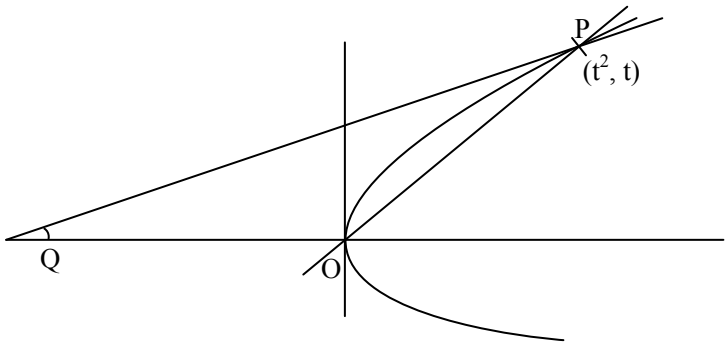
$6 = \frac{2080-81+121}{20} - (10.1)^2$

$= 106 - 102.01 = 3.99$

Q.21 Let a line $y = mx$ $(m > 0)$ intersect the parabola, $y^2 = x$ at a point P, other than the origin. Let the tangent to it at P meet the x-axis at the point Q. If area $(\Delta OPQ) = 4$ sq. units, then m is equal to _____.

Ans. [0.50]

Sol.



Tangent at P $\Rightarrow 2yt = x + t^2$

$Q(-t^2, 0)$

Area $\frac{t^2.t}{2} = 4 \Rightarrow \boxed{t = 2} \Rightarrow P(4, 2)$

P lies on $y = mx$

So $2 = 4m$

$\boxed{m = 1/2}$

Q.22 The number of 4 letter words (with or without meaning) that can be formed from the eleven letters of the word 'EXAMINATION' is _____

Ans. [2454.00]

Sol. 2A, 2I, 2N, E, X, M, T, O

Total words = (all 4 different letters) + (2 alike & 2 different letters) + (2alike & 2 other alike letters)

$^8P_4 + {}^3C_1.{}^7C_2 \frac{4!}{2!} + {}^3C_2 \frac{4!}{2! \times 2!}$

$\frac{8!}{4!} + 3\frac{7.6}{2}\frac{4.3.2}{2} + 3.6$

$= 8.7.6.5 + 63 - 12 + 18$

$= 1680 + 765 + 18 = 2454$

Q.23 The sum, $\sum_{n=1}^{7}\frac{n(n+1)(2n+1)}{4}$ is equal to _____.

Ans. [504.00]

Sol. $\frac{1}{4}\sum_{n=1}^{7}(2n^3+3n^2+n)$

$= \frac{1}{4}\left[2\left(\frac{7.8}{2}\right)^2+3.\frac{7.8.15}{6}+\frac{7.8}{2}\right]$

$= \frac{1}{4}[2.49.16+420+28]$

$= 98.4 + 105 + 7$

$= 504$

Q.24 If $\frac{\sqrt{2}\sin\alpha}{\sqrt{1+\cos 2\alpha}} = \frac{1}{7}$ and $\sqrt{\frac{1-\cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$, $\alpha, \beta \in \left(0,\frac{\pi}{2}\right)$, then $\tan(\alpha+2\beta)$ is equal to _____.

Ans. [1.00]

Sol. $\frac{\sqrt{2}\sin\alpha}{\sqrt{2}\cos\alpha} = \frac{1}{7}$

$\tan\alpha = \frac{1}{7}$

$\tan(\alpha+2\beta) = \frac{\frac{1}{7}+\frac{3}{5}}{1-\frac{1}{7}\frac{3}{4}} = \frac{4+21}{28-3} = \frac{25}{25} = 1$

Q.25 Let f(x) be a polynomial of degree 3 such that f(– 1) = 10, f(1) = – 6, f(x) has a critical point at x = – 1 and f' (x) has a critical point at x = 1. Then f(x) has a local minima at x = _____.

Ans. **[3]**

Sol. $f''(x) = a(x-1)$

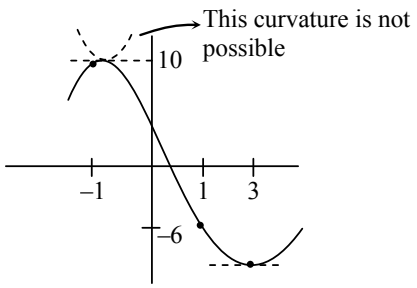
$f'(x) = a\frac{x^2}{2} - ax + b$

$f'(-1) = a/2 + a + b = \frac{3a}{2} + b = 0 \Rightarrow \boxed{-\frac{3a}{2} = b}$

so $f'(x) = a\left(\frac{x^2}{2} - x - \frac{3}{2}\right)$

$= \frac{a}{2}(x^2 - 2x - 3)$

$= \frac{a}{2}(x+1)(x-3)$



So local minima is at x = 3